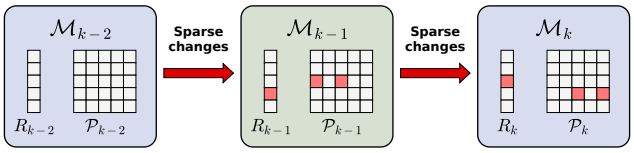
Adaptive Memory for Online Reinforcement Learning under Sparse Model Changes

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Problem and method. Each deployment supplies H observations from a fixed MDP $\mathcal{M}_k = (\mathcal{S}, \mathcal{A}, P_k, R_k, \gamma)$. At each known boundary, at most ϵd reward entries and $\epsilon n d$ transition entries change, with unknown locations and magnitudes. The learner selects separate histories for every reward entry and transition row. It reuses the longest history whose nested confidence sets have a nonempty intersection, fits the model within these intersections, and plans optimally. It stores summaries for at most B completed deployments.



Sparse reward and transition changes across deployments.

Assumptions and notation. Let $n = |\mathcal{S}|$, $d = n|\mathcal{A}|$, and $\vartheta = \min\{1, n\epsilon/2\}$, so at most ϑd transition rows change. Assume $|r| \le \overline{R}$, conditionally independent tuples given an MDP sequence independent of observation noise, sampling coverage $\nu_k(s,a) \ge v > 0$, and discounted occupancies $d^{\pi}_{\rho,k}(s,a) \le \kappa/d$. For discounted return $J^{\rho}_k(\pi)$ from a fixed ρ, define

$$\mathcal{L}^{\rho}_{K,H} := \frac{1}{KH}\sum_{k=0}^{K-1}\sum_{h=0}^{H-1}\left(J^{\rho,*}_k - J^{\rho}_k(\pi_{k,h})\right), \qquad \mathcal{R}^{\rho}_{K,H} := \mathbb{E}[\mathcal{L}^{\rho}_{K,H}],$$
$$L := \log\frac{8d(B+1)(H+1)}{\delta}, \qquad \Lambda := n\log 2 + L, \qquad \beta_{K,B} := K^{-1} + B^{-1}.$$

1. Finite-deployment policy loss. For every $K \ge 1$, integer $B \ge 1$, and $\delta \in (0,1)$,

$$\mathcal{R}^{\rho}_{K,H} \lesssim \frac{\overline{R}}{(1-\gamma)^2}\left[\sqrt{\frac{L(\beta_{K,B}+\kappa\epsilon)}{vH}} + \sqrt{\frac{\Lambda(\beta_{K,B}+\kappa\vartheta)}{vH}}\right] + \frac{\overline{R}\delta}{1-\gamma}. \tag{1}$$

2. High-probability guarantee. Fix K and $\alpha \in (0,1)$, and choose $\delta = \alpha/K$. With $L_\alpha = \log(8dK(B+1)(H+1)/\alpha)$ and $\Lambda_\alpha = n\log 2 + L_\alpha$, with probability at least $1-\alpha$,

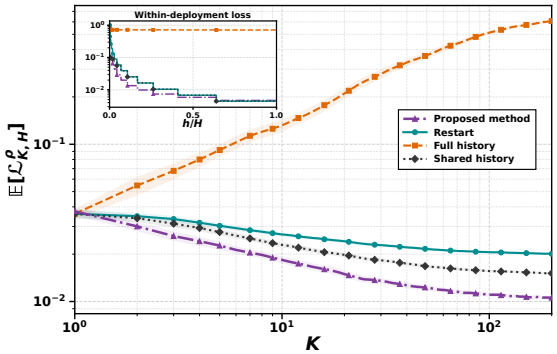
$$\mathcal{L}^{\rho}_{K,H} \lesssim \frac{\overline{R}}{1-\gamma}\sqrt{\frac{L_\alpha(\beta_{K,B}+\kappa\epsilon)}{vH}} + \frac{\gamma\overline{R}}{(1-\gamma)^2}\sqrt{\frac{\Lambda_\alpha(\beta_{K,B}+\kappa\vartheta)}{vH}}. \tag{2}$$

3. Persistent tracking with bounded memory. For $0 < \epsilon \le 1$, choose $B = \lceil \epsilon^{-1} \rceil$ and $\delta = \epsilon/(H+1)$, independently of K. Let $L_\epsilon = \log(24d(H+1)^2/\epsilon^2)$ and $\Lambda_\epsilon = n\log 2 + L_\epsilon$. Then

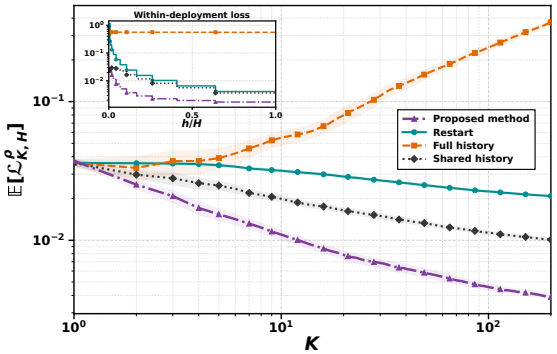
$$\mathcal{J}^{\rho}_H := \limsup_{K\to\infty} \mathcal{R}^{\rho}_{K,H} \lesssim \frac{\overline{R}}{1-\gamma}\sqrt{\frac{\kappa}{vH}}\left[\sqrt{\epsilon L_\epsilon} + \frac{\gamma\sqrt{\vartheta\Lambda_\epsilon}}{1-\gamma}\right]. \tag{3}$$

When $0 < n\epsilon \le 2$, $\mathcal{J}^{\rho}_H = \widetilde{O}(\sqrt{\epsilon/H})$ for fixed remaining parameters. Only $O(\epsilon^{-1})$ deployment summaries are retained, using $O(n^2|\mathcal{A}|/\epsilon)$ scalar storage.

Numerical results. $n = 50$, $|\mathcal{A}| = 4$, $\gamma = 0.8$, $H = 2^{20}$, $K = 200$, and $B = 100$. Results average 50 independent runs. Within each run, all methods receive identical observations and use a common geometric policy-update grid. Reported reductions compare average policy loss over the final 40 deployments against shared history.



(a) $\epsilon = 0.02$: **28.2%** lower policy loss.



(b) $\epsilon = 0.005$: **65.2%** lower policy loss.